

# Pentium II and Pentium III Architecture

## 1 Cache Architecture

Pentium II employs a two-level cache hierarchy to reduce memory access latency and improve overall performance.

### 1.1 Level 1 (L1) Cache

- Total size: **32 KB**
- Divided into:
  - 16 KB Instruction Cache
  - 16 KB Data Cache
- Integrated directly on the processor die
- Provides the fastest access for frequently used instructions and data

### 1.2 Level 2 (L2) Cache

- Implemented using high-speed SRAM
- Located off the processor die but on the same cartridge
- Connected to the CPU using a dedicated **Back-Side Bus (BSB)**
- Typical sizes: 512 KB, 1 MB, or 2 MB

**Exam Note:** Pentium II connects directly to its L2 cache through a dedicated back-side bus rather than directly to main memory.

## 2 Memory System Organization

The Pentium II memory subsystem is designed for high throughput and reliability.

### 2.1 Addressing and Capacity

- Physical address bus width: **36 bits**
- Maximum addressable memory: **64 GB**
- Memory is **byte-addressable**

## 2.2 Memory Width and Type

- Memory data path width: **64 bits (8 bytes)**
- Uses **SDRAM** synchronized with system clock
- Typical access times:
  - 10 ns at 66 MHz
  - 8 ns at 100 MHz

## 2.3 Memory Banks and ECC

- Memory divided into 8 banks (1 byte per bank)
- Optional 9th bank stores **ECC bits**
- ECC provides error detection and correction

**Exam Note:** The 9th memory bank is used exclusively for error-checking code (ECC).

# 3 Chipset and AGP-Based Memory Access

Pentium II introduces a chipset-controlled memory architecture with enhanced graphics support.

- Memory and I/O transfers are controlled by Intel 440LX / 440BX chipsets
- CPU connects directly to L2 cache, not to main memory
- Data transfers between CPU and chipset are 64 bits wide

## 3.1 AGP Memory Mapping

- A portion of main memory is reserved as an **AGP aperture**
- Allows the video card and operating system to access graphics data efficiently
- Results in faster video rendering and updates

### Data Flow:

CPU → L2 Cache → Chipset → Main Memory

# 4 Input / Output System

The Pentium II I/O system preserves full backward compatibility with earlier Intel processors.

- Supports **8192 I/O ports**
- Address lines A15–A3 select the I/O port number

- Address lines A2–A0 select byte, word, or double-word transfers
- Bank-enable signals determine which data banks participate
- All I/O transfers are managed by the chipset

## 5 Pentium III Processor Enhancements

Pentium III is an enhanced version of the Pentium II architecture with improved cache and performance.

### 5.1 Versions

#### 1. External L2 Cache Version

- 512 KB non-blocking L2 cache
- Operates at half of processor speed

#### 2. Advanced Transfer Cache Version

- 256 KB L2 cache integrated on-chip
- Operates at full processor clock speed
- Lower latency and improved efficiency

### 5.2 Additional Features

- Supported chipsets: Intel 810, 815, 820
- System bus speeds: 100 MHz and 133 MHz
- Maximum released speed: 1 GHz
- Packaged in 370-pin PGA (Socket 370)

**Exam Summary:** Pentium III improves performance by integrating faster L2 cache and supporting higher bus and clock speeds.